

Exploratory Data Analysis (EDA) and Machine Learning on Agricultural Yield Dataset

Part A: Understanding the Dataset

Q1. Dataset Overview

Load the dataset and answer the following:

- How many rows and columns are present?
- What are the names of all columns?
- Display the first 10 records.

Q2. Data Types and Missing Values

- Check the data type of each column.
- Identify whether any missing values are present.
- If missing values exist, mention the affected columns.

Q3. Descriptive Statistics

Generate summary statistics for all numerical features and answer:

- Which feature has the highest mean value?
- Which feature has the highest standard deviation?

Part B: Exploratory Data Analysis (EDA)

Q4. Distribution Analysis

Create histograms for:

- rainfall_mm
- temperature_c
- fertilizer_kg
- yield_ton_per_hectare

Write 2–3 observations from each histogram.

Q5. Crop Type Analysis

- Find the number of records for each crop type.
- Create a count plot (bar chart) for crop_type.
- Which crop appears most frequently?

Q6. Soil Type Analysis

- Find the frequency of each soil type.
- Create a count plot for soil_type.
- Which soil type is most common?

Q7. Yield Distribution

Create a histogram of yield_ton_per_hectare.

Answer:

- Is the distribution approximately normal?
- Are there any noticeable outliers?

Q8. Scatter Plot Analysis

Create scatter plots of:

1. rainfall_mm vs yield_ton_per_hectare
2. fertilizer_kg vs yield_ton_per_hectare

Based on the plots:

- Which feature appears to have a stronger relationship with yield?

Q9. Correlation Analysis

- Generate a correlation matrix for numerical features.
- Create a heatmap.
- Identify the top three features most correlated with crop yield.

Q10. Group-Based Analysis

Calculate the average yield for:

- Each crop type
- Each soil type

Which crop and soil type have the highest average yield?

Part C: Data Preparation

Q11. Feature Encoding

The dataset contains categorical variables.

- Identify the categorical columns.
- Convert them into numerical form using One-Hot Encoding.
- Display the first five rows of the transformed dataset.

Q12. Feature Selection

Separate:

- Input features (X)
- Target variable (y)

Specify which column is being used as the target variable.

Part D: Machine Learning

Q13. Train-Test Split

Split the dataset into:

- 80% Training Data
- 20% Testing Data

Display the shape of:

- X_train
- X_test
- y_train
- y_test

Q14. Linear Regression Model

- Train a Linear Regression model.
- Display the model coefficients and intercept.
- Which feature has the highest positive coefficient?